TCU Stats Review Packet

FACT-CHECK REGISTER

Approved public-facing claim register for TCU Stats.

Designed for faster cold review, clearer evidence signaling, and cleaner source verification.

176	14	7
Approved Claims	Sections	Appendix Rows

This register is the verification layer behind TCU Stats. The main body contains only approved rows for public-facing review; internal-only and removed claims are preserved later in a separate appendix.

How To Read This Register

Review flow is intentionally layered: scan the ranked claims first, use trust signals to judge evidence strength, then move into the full register when you need source-level detail.

Review Path	Evidence Labels	Fast Scan
Headline Claims -> Full Register -> Appendix. The main body only includes approved rows.	Support Type tells you how the claim was built. Source Tier tells you what kind of evidence it rests on.	Each section opens with three ranked headline claims before the full table so a reviewer can orient quickly.

Support Types

Display	Meaning
Direct	Directly stated in the cited source or dataset.
Derived	Explicitly calculated from cited source values; the math lives in verification.
Interpretation	Editorial synthesis grounded in cited evidence rather than a direct source sentence.

Source Tiers

Display	Meaning
Primary Paper	Peer-reviewed or primary research source.
Institutional Dataset	Government, institutional, program, or market report source.
Derived Calculation	Reviewer-facing rollup built from approved source values.
Interpretive Synthesis	Framing layer used to connect evidence into reviewer messaging.

Refresh Flag

Refresh	Time-sensitive claim. Recheck before future publication, reuse, or date-sensitive review cycles.

Status Boundary

Status	Meaning	Where It Appears
APPROVED	Cleared for public-facing review and site alignment.	Main body
INTERNAL-ONLY	Useful provenance or context, but not strong enough for public-facing review.	Appendix
REMOVE	Retained for provenance only; not approved for current public-facing use.	Appendix

Section Map

Clickable contents guide. Start with Overview, then move left to right across Carbon & Soil, Economics, and Water.

Overview

Start here for the top-level orientation and strongest cross-site proof points.

Carbon & Soil

Carbon stocks, soil biology, long-term loss, and rebuild pathways.

Economics

Farm-business performance, case studies, and climate-linked financial stress.

Water

Water retention, aquifer decline, Central Valley impacts, and scarcity.

OVERVIEW

14

Approved claims

13 direct | 1 derived

Purpose

Quick-glance headline claims used to orient cold reviewers before they move deeper into the evidence base.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1

133 Gt C global soil carbon lost

Sanderman et al., 2017, PNAS | DOI: 10.1073/pnas.1706103114

Direct
Primary Paper

H2

$500M+/yr annual input losses (US corn fertilizer costs)

Jing et al., 2020, Earth's Future | DOI: 10.1029/2019EF001432

Direct
Primary Paper
Refresh

H3

71% of aquifers are declining

Jasechko & Perrone, 2024, Nature | DOI: 10.1038/s41586-024-07320-8

Direct
Primary Paper
Refresh

Full Register

ID	Claim	Source	Evidence	Review
OV-01	**133 Gt C global soil carbon lost**	Sanderman et al., 2017, PNAS DOI: 10.1073/pnas.1706103114	Direct Primary Paper	Primary-source claim cleared for public review.
OV-02	**$500M+/yr annual input losses (US corn fertilizer costs)**	Jing et al., 2020, Earth's Future DOI: 10.1029/2019EF001432	Direct Primary Paper Refresh	Primary-source claim cleared for public review
OV-03	**71% of aquifers are declining**	Jasechko & Perrone, 2024, Nature DOI: 10.1038/s41586-024-07320-8	Direct Primary Paper Refresh	Primary-source claim cleared for public review
OV-04	**$320B climate disaster losses (2024)**	Munich Re 2024 Annual Report munichre.com	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
OV-05	**0.30–1.00 Mg C/ha/yr annual soil carbon gain across the featured regenerative practices**	Derived from approved rows C4-01 through C4-05 See Rebuilding Carbon rows in this register	Derived Derived Calculation	Derived reviewer claim with explicit math anchored to the cited source values.

ID	Claim	Source	Evidence	Review
OV-06	**+78% higher profits on regen farms**	LaCanne & Lundgren, 2018, PeerJ DOI: 10.7717/peerj.4428	**Direct** **Primary Paper**	Primary-source claim cleared for public review.
OV-07	**+35% water infiltration from cover crops**	Basche & DeLonge, 2019, PLOS ONE DOI: 10.1371/journal.pone.0215702	**Direct** **Primary Paper**	Primary-source claim cleared for public review.
OV-08	**13.12 Gt CO2e/yr moved by mycorrhizal fungi**	Hawkins et al., 2023, Current Biology DOI: 10.1016/j.cub.2023.02.027	**Direct** **Primary Paper** **Refresh**	Primary-source claim cleared for public review
OV-09	**McKinsey: $250B incremental value over a decade (80% adoption)**	McKinsey 2024 McKinsey regenerative agriculture value analysis	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
OV-10	**BCG: 130% long-term ROI on regen transitions**	BCG 2023 BCG long-term ROI analysis	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
OV-11	**USDA: $2.8B committed to Partnerships for Climate-Smart Commodities**	USDA Partnerships for Climate-Smart Commodities, 2023–2024 USDA program summary and award announcements	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
OV-12	**34 years for definitive carbon data (Rodale)**	Rodale Institute FST (1981–2015) rodaleinstitute.org	**Direct** **Institutional Dataset**	Institutional-source claim cleared for public review.
OV-13	**6,000 years for Ogallala to recharge**	USGS 2024 Groundwater Reports USGS Ogallala data	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
OV-14	**50+ years tillage damage on virgin soil (MSU)**	Michigan State University Extension MSU Extension research	**Direct** **Institutional Dataset**	Institutional-source claim cleared for public review.

CARBON POOLS

15

Approved claims

11 direct | 4 derived

Purpose

Defines the scale of soil carbon, where it sits in the global system, and why losses from soil matter to the atmosphere.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1

~2,500 Gt C in global soils (0–2 m)

Direct — Institutional Dataset

Global soil carbon stock estimates; cross-ref IPCC and Sanderman et al., 2017 | Cross-ref global soil carbon stock estimates used in the site carbon-pool visual

H2

~2.8x more carbon in soil than atmosphere

Derived — Derived Calculation

Calculated: 2,500 / 900 Gt C | Soils 2,500 vs Atmosphere 900

H3

Every 1% SOM increase sequesters ~8.9 Gt CO2 globally

Derived — Derived Calculation

Derived from USDA-NRCS + global soil area | Calculation based on global cultivated area

Full Register

ID	Claim	Source	Evidence	Review
C1-01	**~2,500 Gt C in global soils (0–2 m)**	Global soil carbon stock estimates; cross-ref IPCC and Sanderman et al., 2017 Cross-ref global soil carbon stock estimates used in the site carbon-pool visual	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
C1-02	**~2.8x more carbon in soil than atmosphere**	Calculated: 2,500 / 900 Gt C Soils 2,500 vs Atmosphere 900	**Derived** Derived Calculation	Derived reviewer claim with explicit math anchored to the cited source values.
C1-03	**Every 1% SOM increase sequesters ~8.9 Gt CO2 globally**	Derived from USDA-NRCS + global soil area Calculation based on global cultivated area	**Derived** Derived Calculation	Derived reviewer claim with explicit math anchored to the cited source values.
C1-04	**75% of terrestrial carbon stored underground**	Global carbon pool analysis Derived from 2,500 / (2,500 + 560) Gt C	**Derived** Derived Calculation	Derived reviewer claim with explicit math anchored to the cited source values.
C1-06	**Oceans: 38,400 Gt C (90.6%)**	Global carbon cycle estimates IPCC; off-scale reference	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
C1-07	**Soils (0–2 m): 2,500 Gt C (5.9%, largest land pool)**	Sanderman et al., 2017; IPCC DOI: 10.1073/pnas.1706103114	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.

TCU Stats Fact-Check Register

ID	Claim	Source	Evidence	Review
C1-08	**Atmosphere: 900 Gt C (2.1%)**	Global Carbon Budget 2024 globalcarbonproject.org	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
C1-09	**Vegetation: 560 Gt C (1.3%)**	Global carbon cycle estimates IPCC AR6	**Direct** **Institutional Dataset**	Institutional-source claim cleared for public review.
C1-10	**Photosynthesis absorbs 120 Gt C/yr**	Global Carbon Budget globalcarbonproject.org	**Direct** **Institutional Dataset**	Institutional-source claim cleared for public review.
C1-11	**Soil respiration releases 62 Gt C/yr**	Global Carbon Budget globalcarbonproject.org	**Direct** **Institutional Dataset**	Institutional-source claim cleared for public review.
C1-12	**Fossil fuel emissions: 10 Gt C/yr**	Global Carbon Budget 2024 globalcarbonproject.org	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
C1-13	**~6 Gt net atmospheric CO2 accumulation/yr**	Global Carbon Budget 2024 Fossils ~10 + land-use ~1; oceans absorb ~3, land ~3	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
C1-14	**13.12 Gt CO2e channeled by mycorrhizal fungi/yr**	Hawkins et al., 2023, Current Biology DOI: 10.1016/j.cub.2023.02.027	**Direct** **Primary Paper** **Refresh**	Primary-source claim cleared for public review
C1-15	**Equal to 36% of fossil fuel emissions**	Calculated: 13.12 / ~36 Gt CO2 Based on annual fossil CO2	**Derived** **Derived Calculation**	Derived reviewer claim with explicit math anchored to the cited source values.
C1-16	**Mycelial network ~5x Earth-Sun distance (450M km)**	Hawkins et al., 2023 Earth-Sun ~150M km	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review

SOIL MICROBIOME

11

Approved claims
11 direct

Purpose
Supports the biological mechanism layer behind soil carbon storage, fungal transport, and resilience claims.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1
1 billion bacteria per gram of soil
USDA-NRCS soil biology overview; standard soil microbiology reference range | USDA/NRCS soil biology materials
Direct
Institutional Dataset

H2
13.12 Gt CO2e moved by mycorrhizal fungi
Hawkins et al., 2023, Current Biology | DOI: 10.1016/j.cub.2023.02.027
Direct
Primary Paper
Refresh

H3
450M km mycelial network length
Hawkins et al., 2023 | DOI: 10.1016/j.cub.2023.02.027
Direct
Primary Paper
Refresh

Full Register

ID	Claim	Source	Evidence	Review
C2-01	**1 billion bacteria per gram of soil**	USDA-NRCS soil biology overview; standard soil microbiology reference range USDA/NRCS soil biology materials	Direct Institutional Dataset	Institutional-source claim cleared for public review.
C2-02	**13.12 Gt CO2e moved by mycorrhizal fungi**	Hawkins et al., 2023, Current Biology DOI: 10.1016/j.cub.2023.02.027	Direct Primary Paper Refresh	Primary-source claim cleared for public review
C2-03	**450M km mycelial network length**	Hawkins et al., 2023 DOI: 10.1016/j.cub.2023.02.027	Direct Primary Paper Refresh	Primary-source claim cleared for public review
C2-04	**Up to 50% of SOM is microbial necromass**	Lawrence Livermore National Laboratory LLNL microbial necromass research	Direct Institutional Dataset	Institutional-source claim cleared for public review.
C2-05	**Glomalin holds 4–5% of all soil carbon**	Soil science literature Wright & Upadhyaya; glomalin studies	Direct Institutional Dataset	Institutional-source claim cleared for public review.
C2-06	**Glomalin persists 6–42 years**	Soil science literature	Direct	Institutional-source claim cleared for public

ID	Claim	Source	Evidence	Review
		Wright & Upadhyaya	Institutional Dataset	review.
C2-07	**+37% microbial biomass in no-till vs conventional (137 studies)**	Oak Ridge National Laboratory meta-analysis ORNL no-till meta-analysis (137 studies)	**Direct** **Institutional Dataset**	Institutional-source claim cleared for public review.
C2-08	**+45% SOC increase in 8 years (UC Davis, San Joaquin Valley)**	UC Davis 8-year regenerative trial UC Davis San Joaquin Valley trial	**Direct** **Institutional Dataset**	Institutional-source claim cleared for public review.
C2-09	**Each 1% SOM increase stores 20,000 extra gal/acre**	USDA-NRCS USDA-NRCS SOM water holding data	**Direct** **Institutional Dataset**	Institutional-source claim cleared for public review.
C2-10	**Microbes reduce synthetic fertilizer demand 40–50%**	Soil Health Institute; nutrient cycling research SHI 100-farm study data	**Direct** **Institutional Dataset**	Institutional-source claim cleared for public review.
C2-11	**High microbial diversity correlates with drought resilience**	USDA ARS USDA ARS drought resilience research	**Direct** **Institutional Dataset**	Institutional-source claim cleared for public review.

CARBON DEBT - 12,000 YEARS OF LOSS

12

Approved claims

9 direct | 3 derived

Purpose

Documents the historic soil-carbon loss, its CO2 equivalent, and the drawdown opportunity framing.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1

133 Gt C stripped over 12,000 years of human land use

Sanderman et al., 2017, PNAS | DOI: 10.1073/pnas.1706103114

Direct
Primary Paper
Refresh

H2

488 Gt CO2 equivalent

Calculated: 133 Gt C × 3.67 | Standard C-to-CO2 conversion

Derived
Derived Calculation

H3

Largest human-caused biogeochemical disruption

Sanderman et al., 2017 | DOI: 10.1073/pnas.1706103114

Direct
Primary Paper

Full Register

ID	Claim	Source	Evidence	Review
C3-01	**133 Gt C stripped over 12,000 years of human land use**	Sanderman et al., 2017, PNAS DOI: 10.1073/pnas.1706103114	Direct Primary Paper Refresh	Primary-source claim cleared for public review
C3-02	**488 Gt CO2 equivalent**	Calculated: 133 Gt C × 3.67 Standard C-to-CO2 conversion	Derived Derived Calculation	Derived reviewer claim with explicit math anchored to the cited source values.
C3-03	**Largest human-caused biogeochemical disruption**	Sanderman et al., 2017 DOI: 10.1073/pnas.1706103114	Direct Primary Paper	Primary-source claim cleared for public review.
C3-04	**50–70% original carbon gone from cultivated soils**	Lal 2004, Science DOI: 10.1126/science.1097396	Direct Primary Paper	Primary-source claim cleared for public review.
C3-05	**2.5 Mg/ha additional loss 1919–2018 (climate alone)**	Climate-driven SOC loss research Independent of land use decisions	Direct Institutional Dataset	Institutional-source claim cleared for public review.
C3-06	**52% SOC loss: temperate forest to agriculture**	Wei et al., 2014, Scientific Reports DOI: 10.1038/srep05738	Direct Primary Paper	Primary-source claim cleared for public review.
C3-07	**41% SOC loss: tropical forest to agriculture**	Wei et al., 2014	Direct	Primary-source claim cleared for public review.

ID	Claim	Source	Evidence	Review
		DOI: 10.1038/srep05738	Primary Paper	
C3-08	**31% SOC loss: boreal forest to agriculture**	Wei et al., 2014 DOI: 10.1038/srep05738	Direct Primary Paper	Primary-source claim cleared for public review.
C3-09	**42–78 Gt C biologically recoverable**	Lal 2004; Sanderman et al. 2018 Lal: 10.1126/science.1097396; Sanderman: 10.1073/pnas.1706103114	Direct Primary Paper	Primary-source claim cleared for public review.
C3-N01	**42-78 Gt C recoverable = roughly 154-286 Gt CO2 drawdown opportunity.**	Derived from Lal 2004 / Sanderman recovery range 42-78 x 3.67 = 154-286 Gt CO2	Derived Derived Calculation	Derived reviewer claim with explicit math anchored to the cited source values.
C3-N02	**154-286 Gt CO2 equals roughly 4-7 years of current annual CO2 emissions.**	Global Carbon Budget 2024/2025; derived comparison 154-286 Gt CO2 / 41.6 Gt CO2 per year = about 3.7-6.9 years	Derived Derived Calculation Refresh	Derived reviewer claim with explicit math anchored to the cited source values
C3-N03	**Global Carbon Budget 2025: roughly 235 / 585 / 1,110 Gt CO2 remain for 1.5 / 1.7 / 2.0°C**	Global Carbon Budget 2024 / 2025 framing See supplied soil-climate evidence report, Carbon pillar	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review

REBUILDING CARBON

13

Approved claims

11 direct | 2 derived

Purpose

Supports the rates, practices, and long-term trial evidence used to describe the rebuild pathway.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1

Compost (rangeland): 1.00 Mg C/ha/yr (0.8–1.2)

Ryals & Silver, 2013 | Ryals & Silver 2013 compost rangeland

Direct
Institutional Dataset

H2

Cover cropping: 0.56 Mg C/ha/yr (0.32–0.88)

Joshi et al., 2023, Agronomy Journal | Joshi et al. 2023 cover cropping SOC

Direct
Primary Paper
Refresh

H3

AMP grazing: 0.50 Mg C/ha/yr (0.3–2.29)

Multiple grazing studies | Adaptive multi-paddock literature

Direct
Institutional Dataset
Refresh

Full Register

ID	Claim	Source	Evidence	Review
C4-01	**Compost (rangeland): 1.00 Mg C/ha/yr (0.8–1.2)**	Ryals & Silver, 2013 Ryals & Silver 2013 compost rangeland	Direct Institutional Dataset	Institutional-source claim cleared for public review.
C4-02	**Cover cropping: 0.56 Mg C/ha/yr (0.32–0.88)**	Joshi et al., 2023, Agronomy Journal Joshi et al. 2023 cover cropping SOC	Direct Primary Paper Refresh	Primary-source claim cleared for public review
C4-03	**AMP grazing: 0.50 Mg C/ha/yr (0.3–2.29)**	Multiple grazing studies Adaptive multi-paddock literature	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
C4-04	**Organic systems: 0.45 Mg C/ha/yr (0.3–0.6)**	Lal 2004; organic research DOI: 10.1126/science.1097396	Direct Primary Paper Refresh	Primary-source claim cleared for public review
C4-05	**No-till (long-term): 0.30 Mg C/ha/yr (0.1–0.5)**	Conservation tillage literature Long-term aggregate studies	Direct Institutional Dataset	Institutional-source claim cleared for public review.
C4-06	**Rodale 34-yr: Organic 61.9 vs Conventional 53.3 Mg C/ha**	Rodale FST (1981–2015)	Direct	Institutional-source claim cleared for public

ID	Claim	Source	Evidence	Review
		rodaleinstitute.org	Institutional Dataset	review.
C4-07	**+16% more soil carbon (Rodale organic)**	Rodale FST rodaleinstitute.org	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
C4-08	**-45% less energy used (Rodale organic)**	Rodale FST rodaleinstitute.org	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
C4-09	**-40% fewer emissions (Rodale organic)**	Rodale FST rodaleinstitute.org	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
C4-10	**+31% higher drought yields (Rodale organic)**	Rodale FST rodaleinstitute.org	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
C4-11	**Single tillage pass = 50+ years of continuous tillage damage**	MSU Extension, 30-year finding MSU Extension tillage research	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
C4-12	**107M MT CO2e/yr if all US corn adopted cover crops**	US corn acreage × sequestration rates Calculated from USDA data + Joshi et al.	**Derived** Derived Calculation	Derived reviewer claim with explicit math anchored to the cited source values.
C4-N01	**107M MT CO2e per year is roughly equivalent to removing 25 million passenger vehicles from the road.**	EPA greenhouse gas equivalencies; 2024 conversion factors 107,000,000 metric tons / 4.29 metric tons per passenger vehicle = about 24.9 million vehicles	**Derived** Derived Calculation Refresh	Derived reviewer claim with explicit math anchored to the cited source values

PROFITABILITY

16

Approved claims

15 direct | 1 interpretation

Purpose

Validates the core claim that regenerative systems can improve margins while reducing climate-linked input risk.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1
FARM RETURNS
+78% higher profits on regen farms (19 Northern Plains)
LaCanne & Lundgren, 2018, PeerJ | DOI: 10.7717/peerj.4428

Direct
Primary Paper

H2
FARM RETURNS
$51.60 net return/acre above cost — corn
Soil Health Institute 100-farm, 2021 | SHI 2021 100-farm study

Direct
Institutional Dataset
Refresh

H3
CAPITAL SIGNALS
BCG: 130% long-term ROI on regen transitions
BCG 2023 | BCG long-term ROI analysis

Direct
Institutional Dataset
Refresh

Full Register

ID	Claim	Source	Evidence	Review
FARM RETURNS				
E1-01	**+78% higher profits on regen farms (19 Northern Plains)** FARM RETURNS	LaCanne & Lundgren, 2018, PeerJ DOI: 10.7717/peerj.4428	**Direct** Primary Paper	Primary-source claim cleared for public review.
E1-02	**$51.60 net return/acre above cost — corn** FARM RETURNS	Soil Health Institute 100-farm, 2021 SHI 2021 100-farm study	**Direct** Institutional Dataset Refresh	Institutional-source claim cleared for public review
E1-03	**$44.89 net return/acre — soy** FARM RETURNS	Soil Health Institute 100-farm, 2021 SHI 2021	**Direct** Institutional Dataset Refresh	Institutional-source claim cleared for public review
E1-04	**85% of farmers saw positive net returns** FARM RETURNS	SHI 100-farm study SHI 2021	**Direct** Institutional Dataset Refresh	Institutional-source claim cleared for public review

THE EDISON INSTITUTE For Food & Environmental Security

TCU Stats Fact-Check Register

THE CARBON UNDERGROUND

ID	Claim	Source	Evidence	Review
E1-05	**$65/acre average gains (30-farm study)** FARM RETURNS	SHI 30-farm subset SHI dataset	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E1-07	**Input costs: Regen 55 vs Conventional 100 (indexed)** FARM RETURNS	LaCanne & Lundgren; SHI Normalized comparison	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E1-08	**Drought yield: Regen 134 vs Conventional 100** FARM RETURNS	Rodale FST 30-year trial rodaleinstitute.org	Direct Institutional Dataset	Institutional-source claim cleared for public review.
CAPITAL SIGNALS				
E1-10	**McKinsey: $250B incremental value over a decade (80% adoption)** CAPITAL SIGNALS	McKinsey 2024 McKinsey regen agriculture report	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
SYSTEM COSTS				
E1-11	**$500M+/yr hidden fertilizer costs (US corn)** SYSTEM COSTS	Jing et al., 2020 DOI: 10.1029/2019EF001432	Direct Primary Paper Refresh	Primary-source claim cleared for public review
CAPITAL SIGNALS				
E1-13	**BCG: 130% long-term ROI on regen transitions** CAPITAL SIGNALS	BCG 2023 BCG long-term ROI analysis	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
MARKET / PROGRAM SIGNALS				
E1-14	**Carbon credits: $4.50–$11.25/acre/year** MARKET / PROGRAM SIGNALS	Carbon market data Voluntary carbon pricing 2023–2036	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
E1-15	**8M+ acres enrolled (Indigo Ag platform)** MARKET / PROGRAM SIGNALS	Indigo Ag 2023 Platform data	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
E1-16	**2M+ tons CO2e credits verified** MARKET / PROGRAM SIGNALS	Indigo Ag 2023 Platform data	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
E1-17	**USDA climate-smart commodity funding: $2.8B committed** MARKET / PROGRAM SIGNALS	USDA Partnerships for Climate-Smart Commodities, 2023–2024 USDA program summary and award announcements	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
CLIMATE COST FRAMING				
E1-N01	**Nitrous oxide has 273x the 100-year warming power of CO2.** CLIMATE COST FRAMING	IPCC AR6 AR6 WG1 global warming potential values	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E1-N02	**Every dollar of reduced synthetic fertilizer is also a climate metric because lower nitrogen use reduces N2O exposure.** CLIMATE COST FRAMING	IPCC AR6; UNEP Global Nitrous Oxide Assessment 2024 Interpretive synthesis from N2O warming potential and fertilizer emissions literature	Interpretation Interpretive Synthesis Refresh	Interpretive reviewer framing grounded in the cited evidence base

CASE STUDIES

15

Approved claims
15 direct

Purpose
Anchors the argument in named farm examples, platform-scale market signals, and applied business outcomes.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1
BROWN'S RANCH
Brown’s Ranch: SOM 1.9% to 6.1% (+221%)
Gabe Brown / NRCS case study | NRCS documentation

Direct
Institutional Dataset

H2
PLATFORM SIGNALS
Indigo Ag: 8M+ acres enrolled
Indigo Ag 2023 | Platform data

Direct
Institutional Dataset
Refresh

H3
PROGRAM SIGNALS
USDA: $2.8B committed to Partnerships for Climate-Smart Commodities
USDA Partnerships for Climate-Smart Commodities, 2023–2024 | USDA program summary and award announcements

Direct
Institutional Dataset
Refresh

Full Register

ID	Claim	Source	Evidence	Review
BROWN'S RANCH				
E2-01	**Brown’s Ranch: SOM 1.9% to 6.1% (+221%)** BROWN'S RANCH	Gabe Brown / NRCS case study NRCS documentation	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E2-02	**Infiltration 0.5 to 8 in/hr (+16x)** BROWN'S RANCH	Gabe Brown / NRCS NRCS field measurements	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E2-03	**Fertilizer from 100% to -41 to -52%** BROWN'S RANCH	Gabe Brown / NRCS Brown’s Ranch inputs	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E2-04	**Fungicide/herbicide: $0 (100% cut)** BROWN'S RANCH	Gabe Brown / NRCS Brown’s Ranch case study	Direct Institutional Dataset	Institutional-source claim cleared for public review.
WHITE OAK PASTURES				
E2-05	**White Oak Pastures: Revenue <$1M to $20M+ (20x)**	Quantis / WOP LCA, 2019	Direct	Institutional-source claim cleared for public

ID	Claim	Source	Evidence	Review
	WHITE OAK PASTURES	Quantis LCA 2019	Institutional Dataset	review.
E2-06	**Employees 14 to 150+** WHITE OAK PASTURES	Will Harris / WOP WOP public data	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E2-07	**Input costs reduced -80%+** WHITE OAK PASTURES	WOP case study Operational data	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E2-08	**Carbon negative (LCA verified)** WHITE OAK PASTURES	Quantis LCA 2019 Quantis / WOP LCA 2019	Direct Institutional Dataset	Institutional-source claim cleared for public review.
PLATFORM SIGNALS				
E2-09	**Indigo Ag: 8M+ acres enrolled** PLATFORM SIGNALS	Indigo Ag 2023 Platform data	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
E2-10	**2M+ tons CO2e credits verified** PLATFORM SIGNALS	Indigo Ag 2023 Platform data	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
RISK SIGNALS				
E2-11	**$44.4B crop insurance paid 2021–2023** RISK SIGNALS	USDA crop insurance data USDA payout records	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
E2-12	**Record $19.1B paid in 2022** RISK SIGNALS	USDA USDA crop insurance stats	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
CAPITAL SIGNALS				
E2-13	**McKinsey: $250B incremental value over a decade (80% adoption)** CAPITAL SIGNALS	McKinsey 2024 McKinsey regenerative agriculture value analysis	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
E2-14	**BCG: 130% long-term ROI on regen transitions** CAPITAL SIGNALS	BCG 2023 BCG long-term ROI analysis	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
PROGRAM SIGNALS				
E2-15	**USDA: $2.8B committed to Partnerships for Climate-Smart Commodities** PROGRAM SIGNALS	USDA Partnerships for Climate-Smart Commodities, 2023–2024 USDA program summary and award announcements	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review

FARM FINANCIAL STRESS

12

Approved claims

11 direct | 1 derived

Purpose

Shows how climate volatility and debt pressure are already destabilizing conventional farm economics.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1

FARM BALANCE SHEET

315 Chapter 12 farm bankruptcies (2025)

US Courts Chapter 12 | US Courts filing statistics

Direct
Institutional Dataset
Refresh

H2

FARM BALANCE SHEET

$624.7B projected farm debt 2026

USDA ERS | USDA ERS farm debt projections

Direct
Institutional Dataset
Refresh

H3

DISASTER TEMPO

$320B total climate disaster losses 2024

Munich Re 2024 | munichre.com

Direct
Institutional Dataset
Refresh

Full Register

ID	Claim	Source	Evidence	Review
FARM BALANCE SHEET				
E3-01	**315 Chapter 12 farm bankruptcies (2025)** FARM BALANCE SHEET	US Courts Chapter 12 US Courts filing statistics	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
E3-02	**Up 127% from 2023 low** FARM BALANCE SHEET	Calculated: 139 (2023) to 315 US Courts; Am Farm Bureau Fed	Derived Derived Calculation Refresh	Derived reviewer claim with explicit math anchored to the cited source values
E3-03	**$624.7B projected farm debt 2026** FARM BALANCE SHEET	USDA ERS USDA ERS farm debt projections	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
E3-04	**-40% net farm income drop 2022–2024** FARM BALANCE SHEET	USDA ERS USDA ERS farm income data	Direct Institutional Dataset	Institutional-source claim cleared for public review

ID	Claim	Source	Evidence	Review
			Refresh	
E3-05	**140,000+ farms lost 2017–2022** FARM BALANCE SHEET	USDA Census of Agriculture USDA Census	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
E3-06	**+73% farm interest expenses above 2018** FARM BALANCE SHEET	USDA ERS High rates on leveraged operations	Direct Institutional Dataset	Institutional-source claim cleared for public review.
DISASTER TEMPO				
E3-07	**Billion-dollar disasters 1980s: 3.3/yr (82 days)** DISASTER TEMPO	NOAA NOAA Billion-Dollar Disasters	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E3-08	**1990s: 5.5/yr (52 days)** DISASTER TEMPO	NOAA NOAA Billion-Dollar Disasters	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E3-09	**2000s: 6.7/yr (35 days)** DISASTER TEMPO	NOAA NOAA Billion-Dollar Disasters	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E3-10	**2010s: 13.1/yr (24 days)** DISASTER TEMPO	NOAA NOAA Billion-Dollar Disasters	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E3-11	**2020–24: 23/yr (18 days)** DISASTER TEMPO	NOAA NOAA Billion-Dollar Disasters	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
E3-12	**$320B total climate disaster losses 2024** DISASTER TEMPO	Munich Re 2024 munichre.com	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review

CLIMATE & YIELDS

11

Approved claims

11 direct

Purpose

Documents observed yield losses, climate-linked crop pressure, and the resilience case for regenerative management.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1
OBSERVED YIELD LOSSES

Wheat yield: -5.5% (observed, warming)

Lobell et al., 2011, Science | DOI: 10.1126/science.1204531

Direct
Primary Paper
Refresh

H2
OBSERVED YIELD LOSSES

Maize yield: -3.8% (observed, warming)

Lobell et al., 2011 | DOI: 10.1126/science.1204531

Direct
Primary Paper
Refresh

H3
RESILIENCE SIGNAL

Drought: Regen 131 vs Conventional 100 (Rodale)

Rodale FST 30-year | rodaleinstitute.org

Direct
Institutional Dataset

Full Register

ID	Claim	Source	Evidence	Review
OBSERVED YIELD LOSSES				
E4-01	**Wheat yield: -5.5% (observed, warming)** OBSERVED YIELD LOSSES	Lobell et al., 2011, Science DOI: 10.1126/science.1204531	Direct Primary Paper Refresh	Primary-source claim cleared for public review
E4-02	**Maize yield: -3.8% (observed, warming)** OBSERVED YIELD LOSSES	Lobell et al., 2011 DOI: 10.1126/science.1204531	Direct Primary Paper Refresh	Primary-source claim cleared for public review
RESILIENCE SIGNAL				
E4-03	**Drought: Regen 131 vs Conventional 100 (Rodale)** RESILIENCE SIGNAL	Rodale FST 30-year rodaleinstitute.org	Direct Institutional Dataset	Institutional-source claim cleared for public review.
E4-04	**28–34% higher yields during drought** RESILIENCE SIGNAL	Rodale FST Organic corn drought data	Direct Institutional Dataset	Institutional-source claim cleared for public review.

TCU Stats Fact-Check Register

ID	Claim	Source	Evidence	Review
CURRENT MARKET SHOCK				
E4-05	**Cocoa: $12,931/ton (from ~$2,500)** CURRENT MARKET SHOCK	2024 commodity data West Africa crop failure	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
E4-06	**40%+ yield variation from weather extremes** CURRENT MARKET SHOCK	Edison Institute 2024 Edison Institute crop variance research	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
E4-07	**$320B climate disaster losses 2024** CURRENT MARKET SHOCK	Munich Re 2024 munichre.com	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
FUTURE PRESSURE				
E4-08	**-24% maize yield by late century** FUTURE PRESSURE	NASA NASA climate crop modeling	**Direct** **Institutional Dataset**	Institutional-source claim cleared for public review.
E4-09	**+129% cereal price increase by 2050** FUTURE PRESSURE	IPCC IPCC crop/price projections	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
E4-10	**1.4–1.8 pts food inflation/yr by 2035 (N. America)** FUTURE PRESSURE	Kotz et al., 2023 Kotz et al. 2023	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
E4-11	**Every 1°C = $54T kilocalories lost** FUTURE PRESSURE	Hultgren et al., 2025, Nature ~120 kcal/person/day/degree	**Direct** **Primary Paper** **Refresh**	Primary-source claim cleared for public review

SOIL & WATER

13

Approved claims

12 direct | 1 derived

Purpose

Connects soil organic matter to water holding capacity, infiltration, and climate adaptation on the farm.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1

20,000 gal/acre per 1% SOM increase

USDA-NRCS | USDA-NRCS SOM water data

Direct
Institutional Dataset

H2

Plant-available water: 3,400–10,000 gal/ac (Hudson 1994)

Hudson 1994 (Soil Sci. Am. J.) | Hudson 1994

Direct
Institutional Dataset

H3

Infiltration: Perennial +60%

Basche & DeLonge 2019, PLOS ONE (89 studies) | DOI: 10.1371/journal.pone.0215702

Direct
Primary Paper

Full Register

ID	Claim	Source	Evidence	Review
W1-01	**20,000 gal/acre per 1% SOM increase**	USDA-NRCS USDA-NRCS SOM water data	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
W1-02	**Plant-available water: 3,400–10,000 gal/ac (Hudson 1994)**	Hudson 1994 (Soil Sci. Am. J.) Hudson 1994	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
W1-03	**4% SOM holds 2x water of 1% SOM**	Hudson 1994 Hudson 1994	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
W1-04	**Infiltration: Perennial +60%**	Basche & DeLonge 2019, PLOS ONE (89 studies) DOI: 10.1371/journal.pone.0215702	**Direct** Primary Paper	Primary-source claim cleared for public review.
W1-05	**Cover crops +35%**	Basche & DeLonge 2019 DOI: 10.1371/journal.pone.0215702	**Direct** Primary Paper	Primary-source claim cleared for public review.
W1-06	**No-till +45%**	Basche & DeLonge 2019 DOI: 10.1371/journal.pone.0215702	**Direct** Primary Paper	Primary-source claim cleared for public review.
W1-07	**94% erosion reduction: cover crops vs bare soil**	Cover crop studies	**Direct**	Institutional-source claim cleared for public

ID	Claim	Source	Evidence	Review
		USDA-NRCS field data	Institutional Dataset	review.
W1-08	**90% runoff reduction (best-case cover crop)**	Cover crop research Basche & DeLonge meta-analysis	Direct Institutional Dataset	Institutional-source claim cleared for public review.
W1-09	**70% soil loss reduction from ground cover**	IBS studies Institutional field data	Direct Institutional Dataset	Institutional-source claim cleared for public review.
W1-10	**1% SOM=20K gal; 2%=40K (+100%); 4%=80K (+300%)**	Hudson 1994; USDA-NRCS Hudson 1994 data	Direct Institutional Dataset	Institutional-source claim cleared for public review.
W1-11	**Brown's Ranch: 0.5 to 8 in/hr (16x)**	Gabe Brown / NRCS Brown's Ranch case; NRCS	Direct Institutional Dataset	Institutional-source claim cleared for public review.
W1-12	**2T gallons/yr if 1% SOM across 100M US acres**	Calculated: 20K gal × 100M Equiv. 3M Olympic pools	Derived Derived Calculation	Derived reviewer claim with explicit math anchored to the cited source values.
W1-N01	**Each 1°C of warming allows the atmosphere to hold about 7% more water vapor, intensifying both rainfall extremes and drought between them.**	Clausius-Clapeyron relationship; WMO State of Global Water Resources 2024 See supplied soil-climate evidence report, Water pillar, p. 8	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review

AQUIFER CRISIS

11

Approved claims

10 direct | 1 interpretation

Purpose

Supports the groundwater depletion and recharge claims that frame water stress as a climate-linked systems problem.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1

Ogallala: -16.5 ft (1950–2019)

USGS 2024 | USGS Groundwater Reports

Direct
Institutional Dataset
Refresh

H2

Depletion 3–50x faster than recharge

USGS | Recharge ~0.024 in/yr vs pumping ~3 ft/yr

Direct
Institutional Dataset

H3

6,000+ years to replenish naturally

USGS | USGS Groundwater Reports

Direct
Institutional Dataset

Full Register

ID	Claim	Source	Evidence	Review
W2-01	**Ogallala: -16.5 ft (1950–2019)**	USGS 2024 USGS Groundwater Reports	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
W2-02	**Depletion 3–50x faster than recharge**	USGS Recharge ~0.024 in/yr vs pumping ~3 ft/yr	Direct Institutional Dataset	Institutional-source claim cleared for public review.
W2-03	**6,000+ years to replenish naturally**	USGS USGS Groundwater Reports	Direct Institutional Dataset	Institutional-source claim cleared for public review.
W2-04	**Central Valley subsidence: -7.2 ft (2015–2025)**	Stanford 2024 Stanford subsidence research	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
W2-05	**286M acre-feet Ogallala lost since ~1950**	USGS USGS Groundwater Reports	Direct Institutional Dataset	Institutional-source claim cleared for public review.
W2-06	**28 ft max Central Valley subsidence since 1920s**	Historical records USGS / DWR data	Direct Institutional Dataset	Institutional-source claim cleared for public review.

TCU Stats Fact-Check Register

ID	Claim	Source	Evidence	Review
W2-07	**60% Friant-Kern Canal capacity lost**	Stanford Water in the West 2024 Stanford CV subsidence	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
W2-08	**2M AC-FT annual CV groundwater overdraft**	PPIC PPIC Groundwater in CA 2024	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
W2-09	**-1.52 ft Kansas aquifer decline (Jan 24–Jan 25)**	USGS USGS 2024	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
W2-10	**CA State Water Project: 87% decline by 2043**	CA State Water Project 2025 87% delivery loss warning	**Direct** **Institutional Dataset** **Refresh**	Institutional-source claim cleared for public review
W2-N01	**Climate change accelerates aquifer depletion by increasing crop water demand while reducing snowpack recharge.**	Ogallala / Sierra snowpack synthesis in supplied soil-climate evidence report See supplied soil-climate evidence report, Water pillar, pp. 7-8	**Interpretation** **Interpretive Synthesis**	Interpretive reviewer framing grounded in the cited evidence base.

CENTRAL VALLEY

12

Approved claims

11 direct | 1 interpretation

Purpose

Grounds the water crisis in a regional case study with infrastructure, food-system, and land-value consequences.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1
FOOD SYSTEM CONCENTRATION

1% of US farmland

USDA | USDA farmland statistics

Direct
Institutional Dataset

H2
FOOD SYSTEM CONCENTRATION

25% of US food supply

CA agricultural data | USDA / CDFA

Direct
Institutional Dataset

H3
FOOD SYSTEM CONCENTRATION

40% of all US fruits & nuts

CA production data | Almonds, pistachios, grapes

Direct
Institutional Dataset

Full Register

ID	Claim	Source	Evidence	Review
FOOD SYSTEM CONCENTRATION				
W3-01	**1% of US farmland** FOOD SYSTEM CONCENTRATION	USDA USDA farmland statistics	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
W3-02	**25% of US food supply** FOOD SYSTEM CONCENTRATION	CA agricultural data USDA / CDFA	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
W3-03	**40% of all US fruits & nuts** FOOD SYSTEM CONCENTRATION	CA production data Almonds, pistachios, grapes	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
W3-04	**$50B agricultural industry** FOOD SYSTEM CONCENTRATION	CA economic output CA Farm Water Coalition / CDFA	**Direct** Institutional Dataset	Institutional-source claim cleared for public review.
LOCAL ECONOMIC IMPACTS				
W3-05	**Kern County: $132M to $27M = -80% revenue (2023–24)**	Kern County crop data	**Direct**	Institutional-source claim cleared for public

ID	Claim	Source	Evidence	Review
	LOCAL ECONOMIC IMPACTS	CA Farm Water Coalition	Institutional Dataset Refresh	review
W3-06	**17,000 acres permanently removed (2024, SGMA)** LOCAL ECONOMIC IMPACTS	SGMA compliance data First industrial-scale forced fallowing	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
W3-07	**Tulare Lake: 111K acres, $900M–$2B losses** LOCAL ECONOMIC IMPACTS	2023 refilling event Subsidence-damaged drainage	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
W3-08	**Land values: -75% in fallowing zones** LOCAL ECONOMIC IMPACTS	Farmland valuation No water, no crops, collapsing assets	Direct Institutional Dataset	Institutional-source claim cleared for public review.
CROP INTENSITY				
W3-09	**Almonds: 640K to 1,640K acres (+156%)** CROP INTENSITY	CA almond data USDA / Almond Board	Direct Institutional Dataset	Institutional-source claim cleared for public review.
W3-10	**12 liters water per single almond** CROP INTENSITY	Water footprint research 4.7–5.5M ac-ft/yr consumed	Direct Institutional Dataset	Institutional-source claim cleared for public review.
INFRASTRUCTURE STRESS				
W3-11	**Friant-Kern repair: $325M for 10 miles, 74% over budget** INFRASTRUCTURE STRESS	Bureau of Reclamation / DWR Canal lost 60% delivery capacity	Direct Institutional Dataset	Institutional-source claim cleared for public review.
CLIMATE CHAIN				
W3-N01	**Central Valley water stress is a climate-driven chain: warming reduces snowpack, lower deliveries increase pumping, and pumping deepens subsidence.** CLIMATE CHAIN	California snowpack and groundwater synthesis in supplied soil-climate evidence report See supplied soil-climate evidence report, Water pillar, pp. 7-8	Interpretation Interpretive Synthesis	Interpretive reviewer framing grounded in the cited evidence base.

SGMA & FALLOWING

8

Approved claims

7 direct | 1 derived

Purpose

Documents the compliance and land-retirement pressure used to frame regenerative water retention as a strategic alternative.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1

RETIREMENT RISK

500K–1M acres permanently retired by 2040

PPIC projection | PPIC fallowing projections

Direct | Institutional Dataset | Refresh

H2

RETIREMENT RISK

67,000 jobs destroyed (UC Davis worst-case)

UC Davis Economic Analysis 2023 | UC Davis SGMA job loss estimates

Direct | Institutional Dataset | Refresh

H3

RETENTION ALTERNATIVE

Each 1% SOM gain adds roughly 20,000 gal/ac of water-holding capacity

USDA-NRCS | Cross-ref Soil & Water section

Direct | Institutional Dataset

Full Register

ID	Claim	Source	Evidence	Review
POLICY MANDATE				
W4-01	**21 critically overdrafted basins under SGMA** POLICY MANDATE	SGMA / CA DWR DWR basin designations	Direct Institutional Dataset	Institutional-source claim cleared for public review.
W4-02	**2040 sustainability deadline** POLICY MANDATE	SGMA legislation CA SGMA law	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
W4-03	**2M AC-FT annual overdraft to eliminate** POLICY MANDATE	PPIC PPIC Groundwater in CA 2024	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
RETIREMENT RISK				
W4-04	**500K–1M acres permanently retired by 2040** RETIREMENT RISK	PPIC projection PPIC fallowing projections	Direct Institutional Dataset	Institutional-source claim cleared for public review

ID	Claim	Source	Evidence	Review
			Refresh	
W4-05	**5–10% of total CV farmland** RETIREMENT RISK	Calculated from CV acreage ~5–10M total acres	Derived Derived Calculation	Derived reviewer claim with explicit math anchored to the cited source values.
W4-06	**67,000 jobs destroyed (UC Davis worst-case)** RETIREMENT RISK	UC Davis Economic Analysis 2023 UC Davis SGMA job loss estimates	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
W4-07	**$2.5B/yr revenue loss from fallowed land** RETIREMENT RISK	Economic impact estimates Per-acre revenue × fallowed acreage	Direct Institutional Dataset	Institutional-source claim cleared for public review.
RETENTION ALTERNATIVE				
W4-08	**Each 1% SOM gain adds roughly 20,000 gal/ac of water-holding capacity** RETENTION ALTERNATIVE	USDA-NRCS Cross-ref Soil & Water section	Direct Institutional Dataset	Institutional-source claim cleared for public review.

GLOBAL SCARCITY

13

Approved claims

13 direct

Purpose

Extends the water argument from California to the global food-water-climate collision.

Headline Claims

Start here. These are the fastest claims to scan before dropping into the full register.

H1
FRESHWATER DEMAND

70% global ag share of freshwater use

FAO / World Water Resources | FAO global demand modeling

Direct
Institutional Dataset

H2
AQUIFER DECLINE

71% of aquifers declining (2000–2022)

Jasechko & Perrone 2024, Nature | DOI: 10.1038/s41586-024-07320-8

Direct
Primary Paper
Refresh

H3
HUMAN EXPOSURE

5B facing scarcity by 2050

WMO | WMO 5B projection

Direct
Institutional Dataset
Refresh

Full Register

ID	Claim	Source	Evidence	Review
FRESHWATER DEMAND				
W5-01	**70% global ag share of freshwater use** FRESHWATER DEMAND	FAO / World Water Resources FAO global demand modeling	Direct Institutional Dataset	Institutional-source claim cleared for public review.
W5-02	**92% food water footprint from farming** FRESHWATER DEMAND	Water footprint research FAO / WMO	Direct Institutional Dataset	Institutional-source claim cleared for public review.
W5-03	**80% US water consumed by agriculture** FRESHWATER DEMAND	USDA USDA water consumption stats	Direct Institutional Dataset	Institutional-source claim cleared for public review.
AQUIFER DECLINE				
W5-04	**71% of aquifers declining (2000–2022)** AQUIFER DECLINE	Jasechko & Perrone 2024, Nature DOI: 10.1038/s41586-024-07320-8	Direct Primary Paper Refresh	Primary-source claim cleared for public review

TCU Stats Fact-Check Register

ID	Claim	Source	Evidence	Review
W5-05	**71% of 1,700 aquifers dropping** AQUIFER DECLINE	Jasechko & Perrone 2024 DOI: 10.1038/s41586-024-07320-8	Direct Primary Paper Refresh	Primary-source claim cleared for public review
W5-06	**+125% extraction increase in 40 years** AQUIFER DECLINE	Global extraction data Jasechko & Perrone 2024	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
HUMAN EXPOSURE				
W5-07	**4B people affected by water scarcity** HUMAN EXPOSURE	WMO / UN Water WMO Global Water Resources	Direct Institutional Dataset	Institutional-source claim cleared for public review.
AQUIFER DECLINE				
W5-08	**Depletion: 126 (1960s) to 178 (1990s) to 283 km^3/yr (2000s)** AQUIFER DECLINE	Global extraction records Jasechko & Perrone 2024	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
HUMAN EXPOSURE				
W5-09	**5B facing scarcity by 2050** HUMAN EXPOSURE	WMO WMO 5B projection	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
W5-10	**4B face severe scarcity 1+ month/year** HUMAN EXPOSURE	WMO / UN Water Current stress data	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
W5-11	**56% more food calories needed by 2050** HUMAN EXPOSURE	WMO / FAO FAO State of Food & Agriculture	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review
AQUIFER DECLINE				
W5-12	**Critical aquifers: Arabian, Ogallala, Indus, N. China, CV** AQUIFER DECLINE	USGS / global assessments Ranked by depletion severity	Direct Institutional Dataset	Institutional-source claim cleared for public review.
HUMAN EXPOSURE				
W5-13	**2050 gap: food demand vs freshwater = 94 points** HUMAN EXPOSURE	Indexed to 2000 baseline (100) Calories +56, freshwater per capita -38	Direct Institutional Dataset Refresh	Institutional-source claim cleared for public review

Appendix - Internal Provenance

This appendix preserves internal-only and removed rows so provenance is not lost. These pages are not part of the approved cold-review flow.

Internal-Only	Removed	Main Body Rule
3	**4**	**Approved only**
Context kept for internal review, but not cleared for public-facing use.	Rows retained for provenance only and intentionally excluded from current public review.	If a row is not approved, it should appear here rather than in the main section register.

CARBON POOLS

ID	Status	Support	Tier	Claim	Source	Notes
C1-05	REMOVE	Direct	Institutional Dataset	10,000+ microbial species per teaspoon	Soil microbiology consensus	Removed from public-facing materials during synchronization because the species-richness shorthand was broader than the approved evidence base.

CARBON DEBT - 12,000 YEARS OF LOSS

ID	Status	Support	Tier	Claim	Source	Notes
C3-10	REMOVE	Interpretation	Interpretive Synthesis	50–66% recoverable through regen management	Lal 2004; recovery estimates	Removed from public-facing materials during synchronization because the recovery-share shorthand was less defensible than the approved 42–78 Gt C recovery range.

PROFITABILITY

ID	Status	Support	Tier	Claim	Source	Notes
FARM RETURNS						
E1-06	INTERNAL-ONLY	Derived	Derived Calculation	Regen: $1,185/ac vs Conventional: $665/ac net profit	Composite: LaCanne & Lundgren; SHI	Removed from public-facing materials during synchronization; retained only as an internal composite rollup.
E1-09	REMOVE	Derived	Derived Calculation	SOM: Regen 4.2% vs Conventional 2.1% (+100%)	Composite field data	Removed from public-facing materials during synchronization because the paired SOM shorthand was broader than the approved evidence set.
SYSTEM COSTS						
E1-12	REMOVE	Interpretation	Interpretive Synthesis	$1T+/yr broader degradation costs	Neff, CU Boulder, 2021 (informal)	Removed from public-facing materials during synchronization because the broader degradation-cost rollup was not hardened to the same standard as the approved site claims.
FIELD CASE EVIDENCE						
E1-18	INTERNAL-ONLY	Direct	Institutional Dataset	Urnuh: $425/acre income increase over 6 years	TCData 2025	Removed from public-facing materials during synchronization; retained only as internal supporting context.
E1-19	INTERNAL-ONLY	Derived	Derived	ROI: Yr 1–3=30%, Yr 2–5=62%, Yr 3–8=88%, Yr 10+=100%	SHI; LaCanne; Rodale	Removed from public-facing materials

TCU Stats Fact-Check Register

ID	Status	Support	Tier	Claim	Source	Notes
			Calculation		composite	during synchronization; the public site now uses stage labels instead of unsupported percentage milestones.